

Deep-Level Composition Variables as Predictors of Team Performance: A Meta-Analysis

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This study sought to unify the team composition literature by using meta-analytic techniques to estimate the relationships between specified deep-level team composition variables (i.e., personality factors, values, abilities) and team performance. The strength of the team composition variable and team performance relationships was moderated by the study setting (lab or field) and the operationalization of the team composition variable. In lab settings, team minimum and maximum general mental ability and team mean emotional intelligence were related to team performance. Only negligible effects were observed in lab settings for the personality factor and team performance relationships, as well as the value and team performance relationships. In contrast, team minimum agreeableness and team mean conscientiousness, openness to experience, collectivism, and preference for teamwork emerged as strong predictors of team performance in field studies. Results can be used to effectively compose teams in organizations and guide future team composition research.

Keywords: teams and workgroups, team performance, team composition, team design, individual differences

Teams are units of two or more individuals who interact interdependently to achieve a common objective (Baker & Salas, 1997). Teams allow for the completion of tasks that require more than one individual (e.g., decision making, chain customer service). As economic and technological changes continue to place demands on organizations, it is not surprising that teams are now used extensively in organizations (Devine, Clayton, Philips, Dunford, & Melner, 1999). As a complement to the pervasive use of teams in organizations, the performance of teams has been of great interest to researchers. Although early team and group researchers focused mostly on describing group dynamics (e.g., McGrath, 1964; cf. Haythorn, 1953), more recent researchers have focused on how to actively design and manage teams to be more effective (e.g., Hackman, 1987). For example, researchers have focused on manipulating team tasks and the degree to which the team is self-managing (e.g., Langfred, 2004; Man & Lam, 2003) to improve team performance. Another mechanism through which researchers and practitioners have sought to increase team performance is team composition.

Team composition is the configuration of member attributes in a team (Levine & Moreland, 1990) and is thought to have a

powerful influence on team processes and outcomes (Kozlowski & Bell, 2003). Moreland and Levine (1992) suggested that team composition research can be categorized along three dimensions: (a) characteristics of team members (e.g., number of team members, members' abilities, demographics, personality traits), (b) measurement of these characteristics, (c) and the analytical perspective used to approach team composition. The most common analytical perspective views team composition as a cause that shapes various teamwork processes and outcomes of interest. From this perspective, team composition is of interest to both researchers and practitioners as a means of increasing team performance because of the potential ease in manipulating team composition through selection and placement. Consensus on the potential value of team composition has resulted in team composition being one of the most commonly studied team variables (Guzzo & Dickson, 1996; Hollenbeck, DeRue, & Guzzo, 2004). However, despite its popularity with researchers, team composition has been difficult to use because of a lack of understanding in the area (Foushee & Helmreich, 1988; Morgan & Lassiter, 1992). Specifically, it is unclear which specific characteristics and configurations of these characteristics can be used at the design stage of teams to increase team performance.

Team Composition Variables Related to Team Performance

Team performance is defined as the extent to which a team accomplishes its goals or mission (Devine & Philips, 2001). Because team members interact interdependently in order to be successful, team members must engage in a number of team processes or "interdependent acts that convert inputs to outcomes through cognitive, verbal, and behavioral activities directed toward organizing taskwork to achieve collective goals" (Marks, Mathieu, & Zaccaro, 2001, p. 357). Successful performance for any team

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requires members who can complete the technical portions of the team goals or mission (i.e., have specialized expertise), as well as effectively navigate team processes. In general, team composition is thought to be related to team performance because it affects the amount of knowledge and skill team members have to apply to the team task—in terms of both task completion and working interdependently (Hackman, 1987). Team composition may also have implications for the amount of effort team members apply to the task. Consequently, investigations of the team composition variable and team performance relationships have focused on the extent to which certain team member characteristics such as surface- and deep-level attributes in different configurations (e.g., heterogeneity, lowest member on the attribute) are related to engaging in successful team processes as well as team goal attainment.

Surface-level composition variables refer to overt demographic characteristics that can be reasonably estimated after brief exposure, such as age, race, education level, and organizational tenure. Deep-level composition variables refer to underlying psychological characteristics such as personality factors, values, and attitudes. Although research on surface- and deep-level composition variables has evolved somewhat independently, both types of composition variables have received extensive research attention. Much of the research on surface-level composition variables has focused on how demographic heterogeneity may lead to differences in team performance (e.g., Bunderson & Sutcliffe, 2002; Pelled, Eisenhardt, & Xin, 1999). However, empirical studies of the demographic heterogeneity and team performance relationships have been disappointing, and meta-analytic results have failed to demonstrate a relationship between demographic heterogeneity and team performance (Webber & Donahue, 2001). Researchers interested in this area continue to investigate potential demographic heterogeneity and team performance relationships by exploring potential moderators (e.g., Jackson & Joshi, 2004; Jehn & Bezrukova, 2004). Concurrently, researchers have focused on non-demographic, deep-level composition variables both in terms of heterogeneity (Harrison, Price, Gavin, & Florey, 2002) and other team composition variable operationalizations (e.g., Barrick, Stewart, Neubert, & Mount, 1998; Barry & Stewart, 1997). Researchers have suggested that although demographic differences may be important, deep-level composition variables can have a stronger influence on team performance (Harrison et al., 2002; Hollenbeck et al., 2004).

The category of deep-level composition variables includes a range of psychological variables on which people differ, including context-dependent variables (e.g., attitudes) that are highly influenced by manipulations of the team's tasks (e.g., task meaningfulness; Harrison et al., 2002), as well as relatively enduring team member individual differences such as personality factors, values, and abilities. Given the importance of considering the latter at the team design stage (i.e., the potential implications for selection and placement), as well as the potentially strong relationships between deep-level composition variables and team performance, this study focused on relatively enduring deep-level characteristics that might affect performance across teams (i.e., not specialized characteristics needed for a specific type of team).

Although previous empirical studies and meta-analyses have contributed to researchers' understanding of the relationships between enduring deep-level composition variables and team perfor-

mance, this part of the team composition literature remains fragmented and inconsistent, and conclusions regarding optimal composition for high team performance are difficult to make. Specifically, results of empirical studies have been inconsistent regarding which composition variables are predictive of team performance and the relative importance of different composition variables (e.g., Barrick et al., 1998; Mohammed, Mathieu, & Bartlett, 2002), as well as the relationships between different team-level representations of these variables and team performance (e.g., Barrick et al., 1998; Mohammed & Angell, 2003; van Vianen & De Dreu, 2001). Stewart's (2006) meta-analysis on team design and team performance provided some preliminary support for the relationship between "team member personality" and team performance. His results indicated that aggregated personality was related to team performance, but the level of specificity was such that information cannot be garnered on which personality factors, with which operationalizations, are related to team performance. General mental ability (GMA) has emerged as a strong predictor of team performance in two meta-analyses (Devine & Phillips, 2001; Stewart, 2006); however, the extent to which other enduring deep-level composition variables are related to team performance as well as the relative strength of these variables as compared with GMA is unclear. Thus, given the current state of the team composition literature, it is not clear which specific team member characteristics can be used at the design stage of teams to increase team performance and how configurations of team members with these attributes are related to team performance. Consequently, the present study sought to unify the team composition literature by using meta-analytic techniques to estimate the relationships between specified deep-level team composition variables and team performance. Specifically, the objectives of this study were to (a) identify enduring deep-level composition variables studied in the literature that could potentially be used to increase team performance across team types; (b) develop and meta-analytically test hypotheses pertaining to how these variables are related to team performance, as well as the influence of specified moderator variables, including moderators regarding the measurement of the team composition variable; and (c) make theoretically and empirically based recommendations for the optimal composition of teams.

Enduring Deep-Level Composition Variables Related to Team Performance

Literature searches on *team* (or *group*) *performance*, *team* (or *group*) *productivity*, and *team* (or *group*) *effectiveness* were conducted to identify enduring deep-level composition variables related to team performance across team types. Personality factors (using the five-factor model [FFM] of personality; McCrae & Costa, 1987), values (collectivism and preference for teamwork), and abilities (GMA and emotional intelligence [EI]) were identified, reviewed below, and included in the meta-analysis. Although variables beyond those mentioned here have been studied (particularly in the domain of values; e.g., power distance; Earley, 1999) and show promise for future research, they were not included in the review because too few studies were available to justify a meta-analytic integration (number of correlations [*k*] was fewer than five prior to moderator analyses).

Personality Variables

Team member personality is thought to be an important factor in team functioning and performance (e.g., Driskell, Hogan, & Salas, 1987; Hackman, 1987). Much of the progress in personality research in organizations has been attributed to the development of the FFM of personality (McCrae & Costa, 1987), which is a framework for assessing normal personality. The FFM is particularly useful for the integration of numerous studies and was used here to discuss and test the personality variable and team performance relationships. Personality factors of team members are characteristic patterns of thinking, feeling, and acting that should affect team performance through a variety of processes ranging from how team members approach task completion to how team members interact with one another.

Conscientious individuals are described as hardworking, achievement-oriented, and persevering individuals. Conscientiousness has been shown to be related to individual-level performance (Barrick & Mount, 1991; Hurtz & Donovan, 2000). Behaviors associated with conscientious team members should also be beneficial for team performance. For example, conscientious team members engage in behaviors associated with goal completion and problem solving (i.e., enact task roles; Stewart, Fulmer, & Barrick, 2005). This may suggest that conscientiousness should be related to the organization and planning required by many of the transition processes effective teams go through (Marks et al., 2001), such as identifying the team's main tasks and available resources and developing courses of action to complete the team's objectives. Conscientiousness has also been related to backing up behaviors (Porter et al., 2003). Specifically, team members high in conscientiousness were able to identify when they needed assistance from teammates. They secured help when it was needed but did not solicit help from others when they were able to execute their roles without it. Conscientiousness should be related to processes supportive of task completion and goal attainment and thus should be related to team performance.

Several personality factors should be related to how well team members interact with one another interpersonally and interdependently pursue team goals. Individuals who are agreeable (e.g., considerate, trusting, friendly) tend to be better at interpersonal facilitation (Hurtz & Donovan, 2000). Further, individuals high on agreeableness are thought to be motivated to have positive social situations. For example, highly agreeable individuals seek to maintain social harmony and reduce within-group competition (Graziano, Hair, & Finch, 1997). The extent to which a team is composed of agreeable team members may be related to the degree to which team members engage in positive interpersonal processes and ultimately team performance. Higher levels of extraversion are hypothesized to be beneficial when jobs or situations require interpersonal interaction (Barrick & Mount, 1991; Organ & Ryan, 1995). Team member extraversion has been shown to be related to attraction toward the team (Kristof-Brown, Barrick, & Stevens, 2005). In addition to benefiting interpersonal interactions, extraversion has been shown to be related to team processes such as team members seeking help from other team members when needed (Porter et al., 2003). Thus, extraversion may be related to team performance. Individuals who are emotionally stable (e.g., secure, calm, steady) may contribute positively to teamwork and therefore enhance team performance (Hough, 1992; Mount, Bar-

rick, & Stewart, 1998). For example, a team composed of emotionally stable individuals might create a relaxed atmosphere that promotes cooperation and thus might ultimately engage in less disruptive behavior (Reilly, Lynn, & Aronson, 2002).

Finally, individuals who are described as high on openness to experience are considered to be original, imaginative, broad-minded, and daring (McCrae & Costa, 1987). Although openness to experience has been a notoriously weak predictor of individual-level job performance (Barrick & Mount, 1991), it has been suggested that openness to experience is a better predictor when the situation is novel or complex (Griffith & Hesketh, 2004). Openness to experience may be related to team performance to the extent that team members high on this trait are more adaptable and can make the changes required to continue in a dynamic team environment (LePine, 2003).

In summary, the FFM factors of personality should be related to the beneficial processes as suggested above and thus ultimately to team performance. Thus, I made the following hypotheses:

Hypothesis 1: Team conscientiousness will be positively related to team performance.

Hypothesis 2: Team agreeableness will be positively related to team performance.

Hypothesis 3: Team extraversion will be positively related to team performance.

Hypothesis 4: Team emotional stability will be positively related to team performance.

Hypothesis 5: Team openness to experience will be positively related to team performance.

Values

Values are beliefs about desirable behaviors that transcend specific situations, guide the evaluation of behavior, and are ordered in an individual in terms of relative importance (Schwartz & Bilski, 1987). Values are thought to be guides that determine actions, behaviors, and attitudes toward situations. In addition, values are thought to have a strong motivational component (Rokeach, 1973). Values are thought to be relatively enduring, albeit less so than personality factors or intelligence, as they may gradually change over an individual's life span (Rokeach, 1973). The values and team performance relationship has been studied by researchers (e.g., Jehn, 1994; Rodriguez, 1998) using several value taxonomies (e.g., Rokeach, 1973; Schwartz, 1992) as well as value fit in terms of a "team-level" organizational culture profile (O'Reilly, Chatman, & Caldwell, 1991). For many of these approaches to the values and team performance relationship, there is a limited number of studies available in the literature, suggesting it is not yet time for meta-analytic integration. However, the majority of team composition research has investigated the values and team performance relationship in terms of team member collective orientation and preference for teamwork. Specifically, although collectivism was originally thought of in terms of a national-level variable (Hofstede, 2001), teams researchers have focused on its effects within culture and across teams (Kirkman & Shapiro, 2001; labeled allocentrism at the individual level by

Triandis, 2000). Tendencies associated with a collectivistic orientation may be related to team performance across team types. Collectivists tend to prefer procedures that foster harmony and solidarity (Earley & Gibson, 1998). Collectivists also tend to be loyal to their in-group and sacrifice for the sake of the group (Triandis, 1995). Teams composed of team members high on collectivism should foster the cooperation needed in teams (Eby & Dobbins, 1997), and thus, team member collectivism should be related to team performance.

Sometimes considered a facet of collectivism, a second frequently studied concept in the team performance literature is preference for teamwork (e.g., Campion, Medsker, & Higgs, 1993). Preference for teamwork is the degree to which individuals have strong preferences for teamwork rather than autonomous work (Wagner, 1995; Wagner & Moch, 1986). Although research has indicated mixed results regarding a relationship between preference for teamwork and team performance or productivity (e.g., Campion et al., 1993; Jung & Sosik, 1999), for the same reasons outlined above for collectivism, preference for teamwork should be related to team performance. Given that it specifically targets the extent to which team members prefer to work in teams rather than a more general valuing of the in-group, preference for teamwork may be even more strongly related to team performance than the team member collective orientation. Thus, I made the following hypotheses:

Hypothesis 6: Team collectivism will be positively related to team performance.

Hypothesis 7: Team preference for teamwork will be positively related to team performance.

Abilities

In addition to personality variables and values, team member abilities such as GMA should also be related to team performance. Few things have been as widely supported in the industrial and organizational psychology literature as the relationship between GMA and individual-level performance (Schmidt, 2002). A strong relationship between GMA and performance should also be observed in team situations, in which members with high GMA can also develop beneficial team processes related to coordination, such as the development of shared mental models (Edwards, Day, Arthur, & Bell, 2006). Consistent with this are two meta-analyses that have supported the relationship between GMA and team performance (e.g., Devine & Phillips, 2001; Stewart, 2006). Although results from these meta-analyses are available, GMA was coded in the present study to avoid potential problems with comparing results across multiple meta-analyses (Wanous, Sullivan, & Malinak, 1989). Consistent with the extant literature (e.g., Devine & Phillips, 2001), it was expected that GMA would be positively related to team performance.

In addition to the research on GMA, there has been a surge in the study of EI.¹ EI has roots in older concepts such as social intelligence (see Thorndike, 1920; Walker & Foley, 1973) and is offered as a general framework that allows for skills needed to understand and experience emotions most adaptively (Salovey & Mayer, 1990). Specific to the workplace context, EI can be defined as the affective tendencies that enable people to use emotional

information advantageously to achieve desired work outcomes (Mehart, 1998). Although it is noted that questions still remain regarding the construct validity and measurement of EI (Roberts, Zeidner, & Matthews, 2001) or even whether EI is an intelligence or a skill, there is some indication that team EI might be related to team performance.

Observed relationships between EI and task performance at the individual level have been weak (Lam, 1998). These weak relationships may be because the measurement of EI has been characterized by low reliability or because some jobs may be more analytic and autonomous and may require tasks that have little need for EI. Teams may constitute a situation in which EI is needed. Team EI may influence how the team responds to stimuli that elicit emotion (Druskat & Kayes, 1999). Positive emotional reactions are suggested to set the tone for the group and to be related to positive outcomes such as helping behavior (George, 1990). There is some evidence that subscales of EI are differentially related to specified outcomes (e.g., Rapisarda, 2002); however, the limited number of studies on EI and team performance use a variety of measures and subscale dimensions, and thus, only the relationship between overall EI and team performance was tested. Given the potential for emotions to influence teams, and the connection between EI and the management of emotions, EI was expected to be positively related to team performance. Thus, I made the following hypotheses:

Hypothesis 8: Team GMA will be positively related to team performance.

Hypothesis 9: Team EI will be positively related to team performance.

Moderators of the Team Composition Variable and Team Performance Relationships

Specific hypotheses were offered for the specified deep-level composition variable and team performance relationships; however, it was likely that the hypothesized relationships would be moderated by factors related to the method or measurement used by the studies (i.e., operationalization of the team composition variable to the team level, study setting, team tenure). These moderators may help explain the inconsistencies from previous research observed in the team composition variable and team performance relationships. In addition, although the specified team composition variables were thought to be related to team performance regardless of team type, team type was tested as a moderator of the team composition variable and team performance relationships.

Team-Level Operationalization of the Team Composition Variable

Composition variables pose a particular problem, because although individual difference variables are by definition at the

¹ Emotional intelligence is included as an ability for the sake of completeness. Ability is used here as a broad categorization and is not used as a synonym for intelligence or GMA. The debate surrounding the construct validity of EI and its labeling as an intelligence is beyond the scope of the present article.

individual level, the interest in team composition is in the unique combinations of individuals who compose a team (Mohammed et al., 2002), or how the individual-level variables are combined to reflect a team-level operationalization. It is likely that the relationship between team members' composition variables and team performance will be moderated by how the construct is operationalized at the team level, with more appropriate team-level operationalizations of the constructs revealing stronger relationships between the team composition variable and team performance (Arthur, Bell, & Edwards, 2007). Researchers have struggled with this and have used different justifications for choosing among a variety of statistical operationalizations (e.g., mean, variance, minimum, maximum). The most commonly used approaches to choosing an operationalization of the composition variable have been to focus on the nature of the team's task or the nature of the specific trait (Hollenbeck et al., 2004).

Steiner's Typology

Probably the most commonly used approach to specifying the appropriate operationalization has been through the matching of the operationalization to the type of task using Steiner's (1972) task typology (e.g., LePine, Hollenbeck, Ilgen, & Hedlund, 1997; Mohammed et al., 2002; Neuman & Wright, 1999). Because Steiner's task types specifically vary according to the proposed influence of individual members' contribution to performance on the task, this typology is thought to have relevance when unraveling how operationalizations of individual-level variables represent a team-level construct. For additive tasks, the team mean or sum on the variable of interest is thought to be the best operationalization of individual-level variables at the team level, because performance is thought to be equal to the sum of the team's parts. Likewise, the team mean or sum is thought to be the best team-level operationalization of composition variables for teams performing compensatory tasks, because low-performing members' poor performance can be compensated for by high-performing team members. For disjunctive tasks, the team maximum (the best individual within the team on a specified variable) should be the best operationalization of the individual-level phenomenon at the team level, because one member can "solve the problem." For conjunctive tasks, the best team-level representation of contributions of the individual team members should be the team minimum, because the "weakest link" of the team can have a detrimental effect on the performance of such tasks. According to this justification, the relationship between a specified team composition variable and team performance should be moderated by the appropriateness of the match between the task type (using Steiner's task typology) and the operationalization of the team-level construct, such that stronger effects should be observed when an operationalization is appropriate for the task type. Thus, I made the following hypothesis:

Hypothesis 10: The team composition variable and team performance relationships will be stronger when the operationalization of the team composition variable and Steiner's task type match (e.g., team maximum used for disjunctive tasks), compared with when they are mismatched.

Nature of the Trait

Despite the frequency of its use by team composition researchers, Steiner's task typology is not without criticism. Steiner's typology is criticized for being applicable to lab tasks but not to teams occurring in real-world contexts that do multiple tasks. Direct tests of Steiner's typology as a means for specifying a correct operationalization have not been supportive of the approach (e.g., E. A. Day et al., 2004). Instead, the nature of the specific trait may better guide researchers to the best operationalization of the team composition variable.

Kozlowski and Klein (2000) suggested that in efforts to understand multilevel phenomena, researchers should indicate how a lower level phenomenon might manifest itself at a higher level. Kozlowski and Klein suggested that for phenomena that emerge in the same way and are functionally equivalent across levels (e.g., individual and team), appropriate operationalizations are the sum or average. So, for composition variables that are abilities (e.g., GMA) or composition variables that relate to the level of effort the team members apply to the team (e.g., collectivism, preference for teamwork), the best operationalization might be the mean or the sum. In contrast, variables hypothesized to be distinctively different in their structure as they emerge across levels (i.e., individual contributions are not shared or consistent) may require other operationalizations (Kozlowski & Klein, 2000).

Team-level operationalizations of some composition variables may be better represented by operationalizations—such as configural models—that allow for complex patterns of interactions as the team members' individual attributes combine to a team level. Examples of commonly used configural operationalizations are the minimum and the maximum team values, or the team variance (heterogeneity) on the variable of interest. In addition, other researchers have presented alternative configural operationalizations such as creating a "team personality" based on the number of high scores on a specified dimension in a team (Barry & Stewart, 1997). Several hypotheses exist in the literature regarding how certain composition variables might be related to team performance through configural operationalizations. For example, operationalizations such as the minimum or maximum for a team on a composition variable are hypothesized to be important if the characteristic of one team member will have a profound influence on team performance (e.g., one disagreeable member may have the power to disrupt team harmony, leading to decreases in performance). Although research has specifically explored how different operationalizations of team composition variables affect team composition variable and team performance relationships, results have been inconsistent across studies even when multiple operationalizations were used (e.g., Barrick et al., 1998; Neuman, Wagner, & Christiansen, 1999). Meta-analytic procedures should offer some insight into the relationship between specified composition variables and team performance using different operationalizations. Given the potential for operationalization of the team composition variable to moderate the team composition variable and team performance relationships as well as inconsistent results in previous studies, exploratory moderator analyses testing the influence of operationalization were conducted.

Study Setting

Hackman's (1987) original model of how to effectively design and manage teams was conceptualized for teams that exist within the context of an organization. This criterion of organizational context has been reiterated in many team definitions (S. G. Cohen & Bailey, 1997; Kozlowski & Bell, 2003), and some models of team composition include assumptions that team members are working on projects that they believe are relevant to the organization's functioning (e.g., Jackson, May, & Whitney, 1995). However, the control available in lab settings can be important for theory building, and lab settings have been used in the investigation of team composition variable and team performance relationships. Features of the setting (e.g., fidelity) could potentially affect the observed relationships between team composition variables and team performance, especially those composition variables related to performance by fostering beneficial social processes (e.g., personality variables) or how much effort the team members will apply to the task (i.e., values) rather than those related to ability (e.g., GMA). For example, relationships between personality composition variables, values, and team performance may not emerge in artificial lab settings because they require the investment and concern for the outcome more readily experienced by team members in organizational (i.e., field) settings. On the other hand, researchers in industrial/organizational psychology have not typically observed a disparity between research in lab and field settings (Dipboye & Flanagan, 1979). Given its potential influence on the team composition variable and team performance relationships, study setting was investigated as a potential moderator. Thus, I made the following hypothesis:

Hypothesis 11: Personality and value composition variables will have stronger relationships with performance for teams in field settings compared with lab settings.

Team Tenure

Teams usually have a history and a future (Brannick & Prince, 1997), both of which influence current behavior (Hackman, 1992; McGrath, 1990, 1991). This highlights the need to assess teams longitudinally—particularly when history and intrateam relationships have a logical connection to the variable of interest. Composition variables related to effective team processes are thought to be particularly important for teams that do multiple tasks (Cannon-Bowers, Tannenbaum, Salas, & Volpe, 1995). Given that teams are thought to pursue multiple goals over time (Marks et al., 2001), it could be that the strength of the composition variable and team performance relationships increases over time. This may be particularly true for personality factors such as team conscientiousness given that conscientious individuals engage in task role behaviors associated with goal completion (Stewart et al., 2005), and these behaviors may be even more important when teams pursue multiple goals. Thus, team tenure (the amount of time the team has been intact at criterion collection) was tested as a moderator of the team composition variable and team performance relationships. Thus, I made the following hypothesis:

Hypothesis 12: The relationships between personality factors and team performance will be stronger over time.

Team Type

Although the hypothesized variables are thought to be beneficial to team processes and team performance in any type of team, it could be that the relationships between some team composition variables and team performance are dependent on the type of work the team performs. Team type, using Devine's (2002) taxonomy, was investigated as a moderator of the team composition variable and team performance relationships. His taxonomy overcomes some criticisms of other classifications by making categories mutually exclusive and by including more physical team types. Devine specified 12 team types that can be grouped into two larger dimensions, intellectual teams (e.g., advisory, executive) and physical teams (e.g., production, service). Intellectual teams are those in which thinking is fundamental to the task. Mental skills and nonlinear work processes are used in intellectual teams, and new knowledge or information is usually the primary outcome. On the other hand, physical teams are involved with core tasks that require linear processes and physical work skills, with team members applying existing knowledge to create a tangible product. Devine's classification is particularly relevant to the study of team composition because the classification is based on the idea that the determinants of effectiveness vary according to the team context. In his taxonomy, context refers to seven specific task characteristics such as work cycle, temporal duration, task structure, and physical abilities. These task characteristics can be used to generate hypotheses regarding which team situations would allow for stronger team composition variable and team performance relationships. Specifically, GMA might be more important for intellectual teams compared with physical teams. Compared with teams with high task structure (i.e., command, negotiation, production, performance, medical, and transportation), teams with low task structure (i.e., executive, design, and response teams) have to deal with their actions leading to outcomes in a more unpredictable fashion. Openness to experience should be particularly important for low structure teams, given that team members high in openness to experience are more adaptable and can easily make changes in team environments (LePine, 2003). Finally, team maintenance, such as maintaining positive relationships for future interaction for teams, should be more important for teams that have indefinite temporal durations (i.e., executive, production, performance, military) compared with those teams that end on shifts or with each mission (i.e., command, service, medical, response, transportation). Team composition variables such as agreeableness should be more strongly related to team performance in teams with indefinite temporal durations. Given the potential for team type to affect the strength of the team composition variable and team performance relationships, team type was tested as a potential moderator. Because the effect of team tenure was tested, the impact of temporal duration was not tested using team type.

In summary, the main objective was to meta-analytically examine the relationships between specified deep-level composition variables (i.e., personality factors, values, abilities) and team performance. Hypotheses were developed for the specific composition variable and team performance relationships as well as for potential moderators (i.e., operationalization of the team composition variable, study setting, team tenure, and team type).

Method

Literature Search

To locate studies for potential inclusion, *PsycInfo*, *ABI/Inform*, and *ProQuest Digital Dissertations* for the years 1980–April 2006 were searched using keywords such as *team* (or *group*) *performance*; *team* (or *group*) *effectiveness*; and *composition*, *personality*, *abilities*, and *values*, as well as search words associated with specific composition variables once identified for inclusion (e.g., *general mental ability*, *cognitive ability*, *conscientiousness*, *preference for teamwork*). Although initial searches included dates before 1980, the decision to include articles starting with 1980 was made because of (a) very limited composition and performance research before reviews such as those by Hackman (1987) and Gladstein (1984), (b) increased uniformity in the study of personality variables since the development and use of the FFM of personality, and (c) dramatic changes in the description of tasks after this period. The electronic search was supplemented with a manual search of the reference lists from past reviews of the team literature (e.g., S. G. Cohen & Bailey, 1997; Guzzo & Dickson, 1996) and meta-analyses that investigated team composition variables (e.g., Bowers, Pharmer, & Salas, 2000; Devine & Phillips, 2001; Stewart, 2006). In an effort to identify relevant unpublished studies, conference manuals or proceedings for the annual meetings of the Society for Industrial and Organizational Psychology (2003–2006), Academy of Management (2003–2005), and Human Factors and Ergonomics Society (2002–2004) were searched. Abstracts were reviewed for appropriate content (i.e., an investigation of the relationship between one of the specified team composition variables and team performance) and possible inclusion in the meta-analysis. The reference lists of obtained articles were also reviewed to identify additional sources.

Inclusion Criteria

To be included, a study needed to meet the following criteria. First, a study had to investigate the performance of teams. Studies that assessed the relationship between composition variables and performance in groups that were not task interdependent were excluded (e.g., idea generation groups). Studies that investigated teams with children as members and sports teams were not included because they are not likely to be of interest to researchers and practitioners who seek to improve team composition in business-related organizational settings, and because there is some evidence that sports teams behave differently from other teams (Mullen & Cooper, 1994). Second, the performance measure had to reflect the general objective of the team (e.g., not time to completion for a decision-making task). Third, studies had to report sample sizes and the necessary information that allowed for the computation of a correlation that represented the relationship between a team composition variable and team performance at the team level. Because of the inappropriateness of mixing levels of analyses when calculating sample-weighted effects (Beal, Cohen, Burke, & McLendon, 2003), articles that reported only individual-level data were excluded. When correlations were not reported, the appropriate conversion formulas (see Arthur, Bennett, & Huffcutt, 2001, Appendix C.2) were used to convert the reported statistics to correlations. Statistical artifact information (i.e., predictor and criterion reliability) were collected from studies when reported. No

included studies reported enough information for the calculation of range restriction. Finally, only English language articles were included.

Data Set

Applying the specified inclusion criteria resulted in an initial set of correlations from 89 different sources (e.g., journal articles, dissertations). The extent that the correlations were dependent was a function of the specific analyses. An independent data set was constructed for each of the specific moderator analyses. Dependent correlations were represented in the data set by a single linear composite when intercorrelations were reported and by the average of the effects when the intercorrelations were not reported. Attempts were made to avoid redundancies of data (e.g., reanalyses of the same data set; Demko, 2001, and Neuman et al., 1999). Three independent data sets were created, resulting in a data set of 225 independent correlations for the analyses at the level of the specific team composition variables and team performance; a data set of 425 independent correlations for the analyses with the moderators of study setting, team type, and operationalization; and a data set of 286 independent correlations for the analyses with the moderators of operationalization and Steiner's task type. Huffcutt and Arthur's (1995; Arthur et al., 2001) sample-adjusted meta-analytic deviancy statistic was computed to detect outliers. Four potential outliers were identified by the statistic, but a detailed review of the potential outliers revealed no suspicious correlations, so they were not eliminated.

Description of Variables

The type of composition variable was coded (e.g., agreeableness) along with the level of specified moderators, including operationalization of the team composition variable (e.g., mean, minimum) and study setting (i.e., lab or field). Type of team using Devine's (2002) typology (e.g., advisory, design, executive, production) and Steiner's (1972) typology (i.e., additive, disjunctive, conjunctive, compensatory) was coded if the type of team was explicitly stated or described in enough detail to permit a reasonable judgment by the coders. When the description was not sufficient to permit such a judgment, or the type of team was unable to be classified using the specified typologies, type of team was marked "could not be determined." Team tenure was coded as the number of days the team had worked together at criterion data collection.

Coding Accuracy and Interrater Agreement

The author and a post-master's doctoral graduate student served as the coders for the meta-analysis. The author trained the graduate student on the coding scheme. First, he was given a coding sheet along with a sheet that outlined additional information regarding the different variable categorizations. Each coder then independently coded one article. Next, the coders met for a follow-up session to discuss problems encountered in using the coding and information sheets and to make changes to either sheet as deemed necessary. The two coders then coded the same five articles. After coding these five articles, the coders again met, and the degree of interrater agreement between them was assessed. Discrepancies

and disagreements related to the coding of the five articles were resolved using a consensus discussion and agreement between the coders. After the second session, the author and the second coder coded a common set of 20 articles (i.e., 243 correlations) that were used to assess the efficacy of the coding process and decision rules. Interrater agreement was high, with a mean agreement of 99.33% ($SD = 1.75\%$) on the coded variables. Given the high efficacy of the coding process, the author subsequently coded the remaining articles. Articles that were unclear and difficult to code were coded by both coders, and discrepancies were resolved by reaching consensus via discussion.

Meta-Analysis of Correlations

Arthur et al.'s (2001) SAS PROC MEANS meta-analysis program was used to conduct a meta-analysis of correlations using procedures recommended by Hunter and Schmidt (2004). Sample-weighted mean correlations between the team composition variable and team performance were calculated, and the correlations were corrected for unreliability using the artifact distributions for the specified predictor (team composition variable) and team performance. Ninety-five percent confidence intervals were calculated around the sample-weighted mean correlation as a measure of accuracy of the effect size (Whitener, 1990). In addition, the variance accounted for by statistical artifacts, the lower 95% credibility value (95% CV_L), and the standard deviation of the corrected population correlation (SD_p) were used as indicators of

the presence of moderators or that a given effect is dependent on the situation (i.e., there are subpopulations present; Hunter & Schmidt, 2004; Whitener, 1990).

Categorical moderators were assessed with Hunter and Schmidt's (2004) subgroup analysis. In subgroup analysis, a meta-analysis is conducted on each of the specified moderator levels of the relationship of interest. Moderators were inferred to be present if the mean effect sizes estimated in the subgroups differed from the mean across all groups and if the average of the standard deviation of the corrected population correlation of the subgroups was reduced compared with when the groups are collapsed (Hunter & Schmidt, 2004; Whitener, 1990). For the continuous moderator (i.e., team tenure), weighted least squares (WLS) regression was used as suggested by Steel and Kammeyer-Muller (2002). Using such an approach avoids the artificial categorization required by subgroup analysis of moderators that are continuous. For the WLS regressions, the weighting factor was the inverse square root of the sampling error for each correlation (as specified by Steel & Kammeyer-Muller, 2002), and the regression was conducted to see if the moderator explained variance in the uncorrected correlations of the relationships between the team composition variable and team performance.

Results

Results are presented in Tables 1–7 and were interpreted using the rules of thumb for small, medium, and large effect sizes (.10,

Table 1
Meta-Analytic Results for the Relationship Between Conscientiousness and Team Performance

Variable	<i>k</i>	<i>n</i>	<i>SWM r</i>	<i>SWSD</i>	% var. due to sampling error	95% CI		ρ	SD_p	% var. acc. for	95% CV_L
						Lower	Upper				
Conscientiousness ^a	39	2,205	0.09	0.16	69.38	0.04	0.14	0.11	0.11	69.58	−0.07
Lab ^a	28	1,601	0.03	0.13	100.00	−0.02	0.08	0.04	0.00	100.00	0.04
Field ^a	11	604	0.24	0.12	100.00	0.17	0.31	0.30	0.00	100.00	0.30
Mean	32	1,835	0.12	0.18	51.18	0.05	0.18	0.14	0.15	51.35	−0.11
Lab	22	1,310	0.05	0.17	59.77	−0.02	0.12	0.06	0.13	59.81	−0.15
Field	10	525	0.28	0.11	100.00	0.21	0.34	0.33	0.00	100.00	0.33
Maximum	13	702	0.08	0.18	56.57	−0.02	0.18	0.09	0.14	56.56	−0.14
Lab	6	401	0.05	0.18	45.64	−0.09	0.19	0.06	0.16	45.67	−0.20
Field	7	301	0.12	0.17	76.49	−0.01	0.24	0.14	0.10	76.66	−0.03
Minimum	14	794	0.10	0.18	51.96	0.00	0.20	0.12	0.15	52.09	−0.13
Lab	6	417	0.00	0.16	55.24	−0.13	0.12	−0.01	0.13	55.24	−0.22
Field	8	377	0.22	0.12	100.00	0.14	0.30	0.27	0.00	100.00	0.27
Heterogeneity	17	1,133	−0.03	0.18	45.37	−0.12	0.06	−0.03	0.16	45.38	−0.30
Lab	9	688	0.04	0.19	35.55	−0.09	0.16	0.05	0.19	35.57	−0.25
Field	8	445	−0.13	0.10	100.00	−0.20	−0.06	−0.16	0.00	100.00	−0.16
Other	8	437	0.10	0.20	42.05	−0.05	0.24	0.12	0.19	42.14	−0.20
Lab	5	265	−0.02	0.12	100.00	−0.12	0.08	−0.02	0.00	100.00	−0.02
Field	3	172	0.27	0.19	41.22	0.06	0.49	0.33	0.18	42.32	0.04
Steiner's typology											
Matched ^a	17	901	0.11	0.22	39.69	0.00	0.21	0.13	0.21	39.84	−0.21
Not matched ^a	24	1,233	0.02	0.17	69.46	−0.05	0.08	0.02	0.11	69.46	−0.16
Not coded ^a	13	879	0.14	0.15	65.32	0.06	0.22	0.17	0.10	65.91	0.00

Note. Results are corrected for predictor and criterion unreliability. *k* = number of correlations; *n* = number of teams; *SWM r* = sample-weighted mean correlation; *SWSD* = sample-weighted standard deviation of the *SWM r*; % var. due to sampling error = percentage of variance attributed to sampling error; 95% CI = 95% confidence interval; ρ = corrected population correlation; SD_p = standard deviation of the corrected population correlation; % var. acc. for = percentage of variance attributed to sampling error and artifact corrections; 95% CV_L = lower 95% credibility value.

^a To facilitate aggregation, homogeneity was used in the overall effect because the relationship between heterogeneity and performance was negative.

.30, and .50, respectively) suggested by J. Cohen (1992). Given the potential for the theoretical moderators to influence the interpretations of the team composition variable and team performance relationships, summary statements of the influence of the moderators of study setting and operationalization are presented first, followed by tests of the hypothesized team composition variable and team performance relationships.

As indicated in Tables 1–7, study setting was a strong moderator of the team composition variable and team performance relationships. In general, consistent with Hypothesis 11, the relationships between personality factors and team performance and between values and team performance were stronger in field settings compared with lab settings. Small to medium generalizable effects were observed in field settings for all personality and values variables except for emotional stability and collectivism. In lab settings, the relationships between personality factors, values, and team performance were negligible. The reverse effect was observed for ability variables (i.e., GMA, EI) such that the GMA and team performance relationship was stronger in lab settings compared with field settings (see Table 7). The same was observed for the EI and team performance relationships (see Table 6), although the EI relationships were based on a limited number of correlations. In addition, for the relationships between GMA and team performance in both settings, only a small amount of variance was attributed to artifacts, suggesting the presence of additional moderators. The distributions of the correlations are also presented in Figure 1. In lab settings, correlations representing the personality factor and team performance and values and team performance relationships peaked around zero and were fairly evenly distributed. This was not the case for the relationships between personality factors, values, and performance in field settings as well as the relationships between GMA, EI, and team performance in both settings.

Next, the team composition variable and team performance relationships were tested to see if the relationships were stronger when the operationalization matched the task type using Steiner's typology as opposed to mismatched (Hypothesis 10). Contrary to what was hypothesized, results presented in Tables 1–7 indicate that for most composition variables and team performance relationships, similar effects were observed for both when task type using Steiner's typology and the operationalization did not match and when they did. Slightly stronger effects were observed for the conscientiousness, agreeableness, openness, and team performance relationships when task type and operationalization matched than when they did not; however, this may have been driven by the more general influence of the mean operationalization. Stronger effects were observed for EI and collectivism when the task type and operationalizations matched as opposed to mismatched; however, both effects were based on a very limited number of correlations ($k = 2$). Thus, in general, results suggest little support for Hypothesis 10.

Results suggest that the team composition variable and team performance relationships varied as a function of the operationalization of the team composition variable. Thus, given the effect of study setting and operationalization on the team composition variable and team performance relationships, a fully hierarchical moderator analysis was conducted to test the hypothesized team composition variables and team performance relationships. Although in a few cases the estimates for moderators nested within other

moderator levels represented a small number of correlations, the fully hierarchical moderator approach was used to avoid interpretation errors that can result when the influence of combinations of moderators are not tested on a reported relationship (Hunter & Schmidt, 2004). Because there was a limited number of correlations representing alternative operationalizations (e.g., proportion high), these correlations were collapsed into an "other" category. The narrative of the results is limited to the traditional operationalizations (i.e., mean, minimum, maximum, heterogeneity). Alternative operationalizations are referred to in the Discussion section.

Consistent with Hypothesis 1, results presented in Table 1 indicate a positive relationship between team conscientiousness and team performance in field settings ($\rho = .30$; 95% $CV_L = .30$) but no relationship in lab settings ($\rho = .04$; 95% $CV_L = .04$). As suspected, the strength of the relationship was also moderated by operationalization of team conscientiousness. The strongest effect was observed in field settings when team conscientiousness was operationalized as the team mean ($\rho = .33$; 95% $CV_L = .33$) and all variance was attributed to artifacts. This is consistent with the notion that the more conscientious team members are in general, the better the team performance. In lab settings, team conscientiousness was unrelated to team performance regardless of how team conscientiousness was operationalized.

Hypothesis 2 suggested that team agreeableness would be related to team performance. Results indicate a moderate effect for team agreeableness and team performance in field settings ($\rho = .31$; 95% $CV_L = .17$) and again, no effect for team agreeableness and performance in lab settings ($\rho = .03$; 95% $CV_L = .03$). An inspection of the results presented in Table 2 suggests that the operationalization of team agreeableness moderated the team agreeableness and team performance relationship. Consistent with the notion that one disagreeable member can disrupt the social harmony of the team and subsequently, team performance, a moderate relationship between team minimum agreeableness and team performance was observed in field settings ($\rho = .37$; 95% $CV_L = .37$), with all of the variance attributed to artifacts. A similar effect was also observed when team agreeableness was operationalized as the mean ($\rho = .34$; 95% $CV_L = .34$). Team agreeableness was unrelated to team performance regardless of operationalization in lab settings.

Hypothesis 3 suggested that team extraversion would be positively related to team performance. Consistent with this hypothesis, team extraversion was positively related to team performance in field settings ($\rho = .15$; 95% $CV_L = .15$; see Table 3). Team extraversion had a stronger relationship with team performance in field settings when extraversion was operationalized as the team mean ($\rho = .18$; 95% $CV_L = .18$). Inconsistent with the notion that highly extraverted members would be complemented by lower extraverted members, the team extraversion and team performance relationship was negligible in field settings when team extraversion was operationalized as heterogeneity ($\rho = .06$; 95% $CV_L = .06$).

Hypothesis 4 suggested that team emotional stability would be positively related to team performance. Inconsistent with this hypothesis, there were negligible effects in field and lab settings (see Table 4). However, when operationalization was considered, a small to medium effect was observed between team mean emotional stability and team performance in field settings ($\rho = .21$;

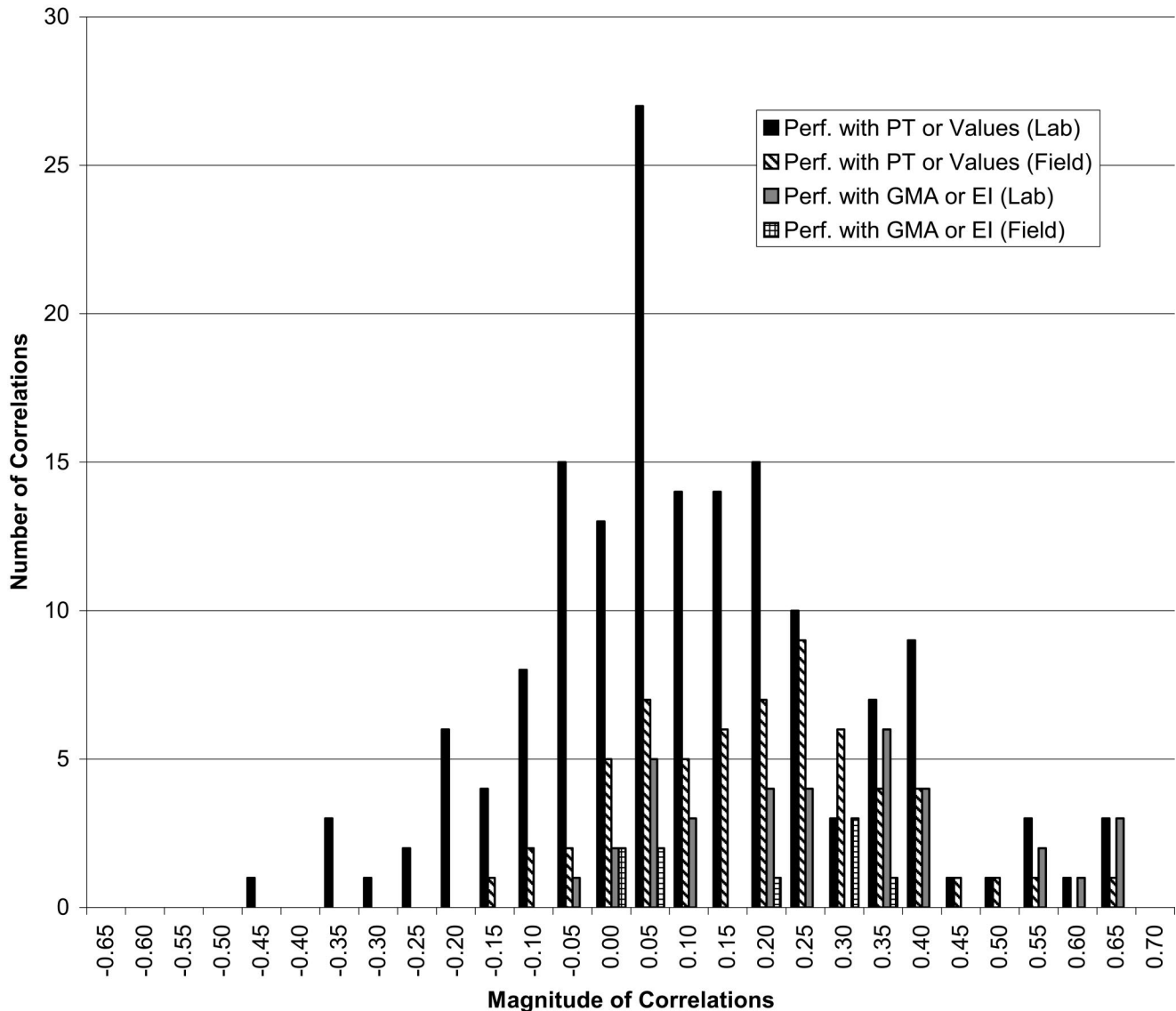


Figure 1. Distribution (histogram) of the 225 correlations included in the meta-analysis representing the relationships between team composition variables and team performance. Perf. with PT or Values = correlations between performance (Perf.) and personality traits (PT) and between performance and values; Perf. with GMA or EI = correlations between performance and general mental ability (GMA) and between performance and emotional intelligence (EI). Values on the x-axis represent the upper value of a .05 band. Thus, for example, the value 0.00 represents correlations falling between $-.05$ and $.00$, and $.30$ represents correlations falling between $.25$ and $.30$.

95% $CV_L = .21$), suggesting that the average emotional stability of team members is positively related to team performance.

Consistent with Hypothesis 5, openness to experience was related to team performance in field settings ($\rho = .20$; 95% $CV_L = .05$), with a fair amount of variance attributed to artifacts (74.93%). Results presented in Table 5 suggest that the relationship was moderated by operationalization of team openness to experience. A small to medium effect was observed for team openness to experience and team performance when openness was operationalized as the team mean ($\rho = .25$; 95% $CV_L = .16$).

Results for Hypotheses 6–7 (collectivism, preference for teamwork) are presented in Table 6. Hypothesis 6 suggested that collectivism would be positively related to team performance. Only 14 correlations represented the effect, with the majority in lab settings ($k = 9$). There was a medium to large effect for collectivism and team performance in field settings when collectivism was operationalized as the team mean ($\rho = .40$; 95% $CV_L = 0$); however, only a very limited amount of variance was attributed to artifacts (12.09%). This suggests that although there is potential for a fairly strong positive relationship between collectivism and

Table 2
Meta-Analytic Results for the Relationship Between Team Agreeableness and Team Performance

Variable	<i>k</i>	<i>n</i>	<i>SWM r</i>	<i>SWSD</i>	% var. due to sampling error	95% CI		ρ	SD_{ρ}	% var. acc. for	95% CV_L
						Lower	Upper				
Agreeableness ^a	29	1,692	0.10	0.17	58.69	0.04	0.16	0.12	0.13	58.93	-0.10
Lab ^a	19	1,118	0.02	0.13	100.00	-0.03	0.08	0.03	0.00	100.00	0.03
Field ^a	10	574	0.25	0.14	75.81	0.16	0.34	0.31	0.08	78.00	0.17
Mean	23	1,303	0.13	0.18	54.28	0.06	0.21	0.17	0.15	54.54	-0.08
Lab	14	808	0.05	0.16	67.81	-0.04	0.13	0.06	0.11	67.84	-0.13
Field	9	495	0.28	0.10	100.00	0.21	0.34	0.34	0.00	100.00	0.34
Maximum	10	482	0.07	0.18	65.23	-0.04	0.18	0.09	0.13	65.30	-0.12
Lab	3	181	0.01	0.07	100.00	-0.07	0.08	0.01	0.00	100.00	0.01
Field	7	301	0.11	0.21	52.11	-0.04	0.27	0.14	0.18	52.22	-0.16
Minimum	13	694	0.15	0.21	41.85	0.04	0.27	0.19	0.20	42.10	-0.13
Lab	5	317	-0.02	0.14	84.27	-0.14	0.10	-0.02	0.07	84.29	-0.14
Field	8	377	0.30	0.13	100.00	0.21	0.39	0.37	0.00	100.00	0.37
Heterogeneity	15	912	-0.03	0.13	100.00	-0.09	0.03	-0.04	0.00	100.00	-0.04
Lab	7	467	0.01	0.11	100.00	-0.07	0.09	0.01	0.00	100.00	0.01
Field	8	445	-0.07	0.13	100.00	-0.15	0.02	-0.08	0.00	100.00	-0.08
Other	6	346	0.12	0.20	42.11	-0.04	0.28	0.15	0.19	42.28	-0.16
Lab	3	174	-0.05	0.11	100.00	-0.18	0.07	-0.07	0.00	100.00	-0.07
Field	3	172	0.30	0.09	100.00	0.20	0.40	0.36	0.00	100.00	0.36
Steiner's typology											
Matched ^a	15	853	0.15	0.21	39.83	0.04	0.25	0.18	0.20	40.11	-0.14
Not matched ^a	17	800	0.09	0.16	80.22	0.01	0.17	0.11	0.09	80.38	-0.04
Not coded ^a	7	546	0.08	0.15	53.72	-0.04	0.19	0.09	0.13	53.90	-0.12

Note. Results are corrected for predictor and criterion unreliability. *k* = number of correlations; *n* = number of teams; *SWM r* = sample-weighted mean correlation; *SWSD* = sample-weighted standard deviation of the *SWM r*; % var. due to sampling error = percentage of variance attributed to sampling error; 95% CI = 95% confidence interval; ρ = corrected population correlation; SD_{ρ} = standard deviation of the corrected population correlation; % var. acc. for = percentage of variance attributed to sampling error and artifact corrections; 95% CV_L = lower 95% credibility value.

^a To facilitate aggregation, homogeneity was used in the overall effect because the relationship between heterogeneity and performance was negative.

team performance, additional moderators beyond those studied here seem to be affecting the strength of the relationship. Hypothesis 7 suggested a positive relationship between preference for teamwork and team performance. Consistent with this, a small to medium effect was observed for the preference for teamwork and team performance relationship in field settings ($\rho = .22$; 95% $CV_L = .22$), with all variance attributed to artifacts.

Hypothesis 8 suggested a positive relationship between team GMA and team performance. Results presented in Table 7 indicate that the mean was generally used to operationalize team GMA in both settings. Despite the presence of small to medium effects between team mean GMA and team performance in lab and field settings ($\rho = .33$; 95% $CV_L = -.01$ and $\rho = .26$; 95% $CV_L = -.06$, respectively), only a small amount of variance was attributed to artifacts, suggesting variability in estimates across situations. In contrast, for the personality factor and team performance relationships, most of the variance was attributed to artifacts when the operationalization of the team composition variable and study setting were considered.

Hypothesis 9 suggested that team EI would be positively related to team performance. Results presented in Table 7 suggest that the relationship was represented by few correlations ($k = 6$), all of which operationalized team EI as the mean and the majority of which investigated the relationship in lab settings ($k = 4$). Results indicate a small to medium effect in lab settings, with all variance attributed to artifacts ($\rho = .20$; 95% $CV_L = .20$); a smaller effect

was observed in field settings, but the estimate was based on only two correlations.

Finally, team tenure and team type were tested as moderators of the personality factor and team performance relationships. Hypothesis 12 suggested that the relationships between the personality factors and team performance would be stronger the longer the team had been together. Results from all WLS regressions were not statistically significant, suggesting that team tenure did not account for a significant amount of variance in the uncorrected correlations representing the personality factor and team performance effects. Low power was most likely the reason for not obtaining statistically significant effects. The sample size for these analyses was very limited because a separate regression was conducted for each composition variable and only some studies reported team tenure. Descriptively, team tenure explained 11% of the variance in the team conscientiousness and team performance relationships ($R^2 = .11$), $F(1, 22) = 2.76$, $p = .11$, and 12% of the variability in the emotional stability and team performance relationships ($R^2 = .12$), $F(1, 9) = 1.21$, $p = .30$. All other analyses testing the influence of team tenure on the relationships between personality factors and team performance accounted for less than 6% of the variation in the effect sizes.

Analyses were conducted to test if the team composition variable and team performance relationships were stronger for team contexts in which the team composition variable was thought to be particularly relevant. First, team GMA was suspected to have a

Table 3
Meta-Analytic Results for the Relationship Between Team Extraversion and Team Performance

Variable	<i>k</i>	<i>n</i>	<i>SWM r</i>	<i>SWSD</i>	% var. due to sampling error	95% CI		ρ	SD_{ρ}	% var. acc. for	95% CV_L
						Lower	Upper				
Extraversion	38	2,243	0.07	0.13	94.81	0.03	0.11	0.09	0.04	95.01	0.03
Lab	29	1,590	0.05	0.14	94.19	0.00	0.10	0.06	0.04	94.28	-0.01
Field	9	653	0.12	0.10	100.00	0.06	0.19	0.15	0.00	100.00	0.15
Mean	27	1,486	0.08	0.14	95.02	0.03	0.13	0.10	0.04	95.20	0.04
Lab	20	994	0.05	0.15	91.49	-0.01	0.12	0.06	0.05	91.55	-0.02
Field	7	492	0.14	0.08	100.00	0.08	0.20	0.18	0.00	100.00	0.18
Maximum	8	408	0.09	0.14	97.82	-0.01	0.18	0.11	0.02	97.99	0.07
Lab	3	181	0.06	0.10	100.00	-0.05	0.17	0.08	0.00	100.00	0.08
Field	5	227	0.11	0.17	79.87	-0.04	0.25	0.13	0.09	80.01	-0.02
Minimum	10	573	0.04	0.13	100.00	-0.04	0.12	0.05	0.00	100.00	0.05
Lab	5	317	0.04	0.10	100.00	-0.05	0.13	0.05	0.00	100.00	0.05
Field	5	256	0.04	0.15	86.11	-0.10	0.17	0.04	0.07	86.12	-0.07
Heterogeneity	19	1,207	0.03	0.10	100.00	-0.02	0.07	0.03	0.00	100.00	0.03
Lab	14	883	0.02	0.10	100.00	-0.04	0.07	0.02	0.00	100.00	0.02
Field	5	324	0.05	0.09	100.00	-0.03	0.13	0.06	0.00	100.00	0.06
Other	8	515	0.04	0.15	66.45	-0.07	0.15	0.05	0.11	66.48	-0.13
Lab	6	383	-0.01	0.15	68.70	-0.13	0.11	-0.01	0.10	68.70	-0.18
Field	2	132	0.18	0.00	100.00	0.17	0.18	0.21	0.00	100.00	0.21
Steiner's typology											
Matched	14	778	0.04	0.16	71.79	-0.04	0.12	0.05	0.10	71.83	-0.12
Not matched	23	1,245	0.05	0.13	100.00	-0.01	0.10	0.06	0.00	100.00	0.06
Not coded	12	817	0.08	0.10	100.00	0.02	0.13	0.09	0.00	100.00	0.09

Note. Results are corrected for predictor and criterion unreliability. *k* = number of correlations; *n* = number of teams; *SWM r* = sample-weighted mean correlation; *SWSD* = sample-weighted standard deviation of the *SWM r*; % var. due to sampling error = percentage of variance attributed to sampling error; 95% CI = 95% confidence interval; ρ = corrected population correlation; SD_{ρ} = standard deviation of the corrected population correlation; % var. acc. for = percentage of variance attributed to sampling error and artifact corrections; 95% CV_L = lower 95% credibility value.

stronger relationship with team performance in intellectual teams compared with physical teams. Results are presented in Table 7. There was substantial overlap among study setting and type of team, such that nearly all studies in lab settings assessed the GMA and performance relationship in intellectual teams. The reverse was true in field settings, such that nearly all studies assessed the GMA and performance relations in physical teams. The limited number of correlations in each cell suggests that GMA is related to team performance in both physical and intellectual teams. The relationship between team openness to experience and team performance was thought to be stronger for teams with low structure compared with high structure. There was a limited number of correlations from low structure teams (see Table 5); however, these few correlations do not suggest a stronger relationship for team openness to experience and team performance in low structure teams.

Discussion

Overall, several deep-level composition variables were related to team performance. In field settings, many of the team composition variable and team performance relationships were at least a medium effect (e.g., team minimum agreeableness $\rho = .37$; team mean conscientiousness $\rho = .33$) or small effect (e.g., team mean emotional stability $\rho = .21$; team mean preference for teamwork $\rho = .26$). Several deep-level composition variable and team performance relationships were notably larger than or similar to

meta-analytic effects observed for team process and performance relationships, such as the team cohesion and team performance outcome relationship ($\rho = .17$; Beal et al., 2003) and the team conflict and team performance relationship (task conflict $\rho = -.23$; relationship conflict $\rho = -.22$; De Dreu & Weingart, 2003). Several of the team-level effects were stronger compared with those typically observed at the individual level (e.g., a $\rho = .12$ for agreeableness and individual-level job performance, a $\rho = .24$ for conscientiousness and individual-level job performance; Hurtz & Donovan, 2000). Although designing the composition of teams may not be practical in every situation, the results here suggest that when feasible (e.g., organizational restructuring, selection for team-based jobs, selection into teams), researchers and practitioners can use team composition to increase team performance.

Two consistent findings for the team composition variable and team performance relationships were observed. Specifically, the relationships were strongly moderated by the setting in which the studies were conducted and the team-level operationalization of the composition variable. Although results indicate that personality factors and collectivism were related to team performance in field settings, there was generally no effect for these variables in lab settings. This does not necessarily mean that lab research studying the deep-level composition variable and team performance relationships should be abandoned. First, lab studies can be designed in such a way to help understand the mechanisms through which the team composition variables are related to performance (e.g.,

Table 4
Meta-Analytic Results for the Relationship Between Team Emotional Stability and Team Performance

Variable	<i>k</i>	<i>n</i>	<i>SWM r</i>	<i>SWSD</i>	% var. due to sampling error	95% CI		ρ	<i>SD_ρ</i>	% var. acc. for	95% <i>CV_L</i>
						Lower	Upper				
Emotional stability	22	1,439	0.03	0.13	93.50	-0.02	0.09	0.04	0.04	93.54	-0.02
Lab	16	1,006	0.03	0.13	96.02	-0.04	0.09	0.03	0.03	96.04	-0.02
Field	6	433	0.05	0.12	89.68	-0.05	0.15	0.06	0.05	89.81	-0.02
Mean	17	1,125	0.11	0.11	100.00	0.05	0.16	0.13	0.00	100.00	0.13
Lab	12	771	0.07	0.11	100.00	0.01	0.14	0.09	0.00	100.00	0.09
Field	5	354	0.18	0.09	100.00	0.10	0.26	0.21	0.00	100.00	0.21
Maximum	7	388	0.11	0.19	49.62	-0.03	0.25	0.13	0.16	49.75	-0.13
Lab	3	181	0.12	0.14	79.66	-0.05	0.28	0.14	0.08	79.91	0.01
Field	4	207	0.11	0.22	38.77	-0.11	0.33	0.13	0.21	38.83	-0.22
Minimum	9	575	0.05	0.11	100.00	-0.02	0.13	0.07	0.00	100.00	0.07
Lab	5	339	0.09	0.08	100.00	0.02	0.16	0.11	0.00	100.00	0.11
Field	4	236	0.00	0.13	100.00	-0.12	0.12	0.00	0.00	100.00	0.00
Heterogeneity	12	855	0.01	0.16	87.23	-0.07	0.10	0.02	0.12	57.24	-0.18
Lab	8	551	0.05	0.12	100.00	-0.03	0.13	0.06	0.00	100.00	0.06
Field	4	304	-0.04	0.20	33.24	-0.24	0.15	-0.05	0.20	33.26	-0.37
Other	5	306	-0.01	0.22	35.29	-0.20	0.18	-0.01	0.21	35.29	-0.35
Lab	3	174	-0.18	0.09	100.00	-0.28	-0.08	-0.21	0.00	100.00	-0.21
Field	2	132	0.21	0.11	100.00	0.06	0.37	0.26	0.00	100.00	0.26
Steiner's typology											
Matched	11	672	0.06	0.12	100.00	-0.01	0.12	0.07	0.00	100.00	0.07
Not matched	13	778	0.05	0.12	100.00	-0.02	0.12	0.06	0.00	100.00	0.06
Not coded	6	499	0.00	0.12	91.02	-0.10	0.09	-0.01	0.04	91.03	-0.07

Note. Results are corrected for predictor and criterion unreliability. *k* = number of correlations; *n* = number of teams; *SWM r* = sample-weighted mean correlation; *SWSD* = sample-weighted standard deviation of the *SWM r*; % var. due to sampling error = percentage of variance attributed to sampling error; 95% CI = 95% confidence interval; ρ = corrected population correlation; *SD_ρ* = standard deviation of the corrected population correlation; % var. acc. for = percentage of variance attributed to sampling error and artifact corrections; 95% *CV_L* = lower 95% credibility value.

Graziano et al., 1997). Second, additional analyses revealed a striking bias in the types of teams that were studied in each setting, such that the majority of lab studies (75.46%) assessed the performance of intellectual teams (e.g., command and control, design), whereas the majority of field studies (75.81%) assessed the performance of physical teams (e.g., service, production). Thus, although it is clear there is a moderator (or moderators) affecting the team composition variable and team performance relationships, it is unclear whether it is study setting, team type, or both. In addition, there seems to be some variability in the relationships across specific lab protocols. Specifically, supplemental analyses suggested there was a small team conscientiousness and team performance relationship in classroom projects ($\rho = .14$; 95% *CV_L* = .14), a small team emotional stability and team performance relationship in computer-based lab protocols ($\rho = .15$; 95% *CV_L* = .15), and a small team extraversion and team performance relationship in computer-based lab protocols ($\rho = .10$; 95% *CV_L* = .10). All other personality factor and team performance relationships were negligible regardless of lab protocol used. Third, although team personality factor and team performance relationships were negligible in lab settings, this was not the case for the EI and team performance relationship or the GMA and team performance relationship. Given the results of the meta-analysis, observed team composition variable and team performance relationships should be discussed within the appropriate boundary conditions. The potential influence of certain deep-level composition variables should not be arbitrarily disregarded if nonsignificant findings are observed in lab settings.

The second consistent finding was that the composition variable and team performance relationships were moderated by the operationalization of the composition variable to the team level. Despite the popularity of Steiner's task typology with team researchers, results from the meta-analysis offer little support for the application of Steiner's task typology as a justification for identifying the most appropriate operationalization of the team composition variable. Further, consistent with the notion that Steiner's typology is less applicable in field settings (e.g., the notion that teams manage bundles of activities rather than one specific task; Marks et al., 2001), many of the articles either did not report enough information or could not be coded into the typology. The difficulty in describing real-world teams with the typology, compounded with the lack of strong relationships when using an appropriate match, calls into question the continued use of Steiner's typology to specify the best operationalization of composition variables at the team level.

Further, no single operationalization was best for all composition variables; rather, the best operationalization was dependent on the specific team composition variable of interest. For agreeableness in field settings and GMA in lab settings, the strongest relationships with team performance were observed when the composition variable was operationalized as the team minimum. However, most of the composition variable and team performance relationships were strongest when the composition variable was operationalized as the mean. This may be because measures of central tendency (e.g., mean) are the single best representation of a distribution (the team scores on the composition variable). A

Table 5
Meta-Analytic Results for the Relationship Between Team Openness to Experience and Team Performance

Variable	<i>k</i>	<i>n</i>	<i>SWM r</i>	<i>SWSD</i>	% var. due to sampling error	95% CI		ρ	SD_{ρ}	% var. acc. for	95% CV_L
						Lower	Upper				
Openness to experience ^a	25	1,697	0.04	0.16	58.02	-0.02	0.10	0.05	0.13	58.07	-0.17
Lab ^a	19	1,295	0.00	0.15	68.33	-0.06	0.07	0.00	0.10	68.33	-0.17
Low structure ^a	6	301	-0.01	0.18	61.69	-0.16	0.13	-0.02	0.14	61.69	-0.28
High structure ^a	4	225	-0.06	0.19	51.53	-0.24	0.12	-0.07	0.16	51.62	-0.33
Field ^a	6	402	0.16	0.14	73.86	0.05	0.27	0.20	0.09	74.93	0.05
Low structure ^a	1	147	0.16								
Mean	20	1,383	0.09	0.16	59.34	0.02	0.16	0.11	0.12	59.52	-0.09
Lab	15	1,060	0.05	0.15	66.70	-0.02	0.13	0.07	0.10	66.78	-0.11
Field	5	323	0.20	0.13	85.39	0.09	0.32	0.25	0.06	86.88	0.16
Maximum	10	629	0.08	0.11	100.00	0.01	0.15	0.10	0.00	100.00	0.09
Lab	6	453	0.06	0.11	100.00	-0.03	0.14	0.07	0.00	100.00	0.07
Field	4	176	0.14	0.11	100.00	0.03	0.24	0.17	0.00	100.00	0.17
Minimum	10	667	0.04	0.17	55.67	-0.06	0.15	0.05	0.14	55.71	-0.17
Lab	6	462	0.03	0.18	40.52	-0.12	0.17	0.04	0.17	40.53	-0.24
Field	4	205	0.07	0.12	100.00	-0.04	0.19	0.09	0.00	100.00	0.09
Heterogeneity	10	681	-0.02	0.13	87.12	-0.10	0.06	-0.03	0.06	87.14	-0.12
Lab	6	408	0.03	0.13	84.54	-0.07	0.14	0.04	0.06	84.56	-0.06
Field	4	273	-0.11	0.07	100.00	-0.17	-0.04	-0.13	0.00	100.00	-0.13
Other	5	306	0.03	0.26	25.10	-0.19	0.26	0.04	0.27	25.11	-0.41
Lab	3	174	-0.18	0.10	100.00	-0.29	-0.07	-0.22	0.00	100.00	-0.22
Field	2	132	0.31	0.03	100.00	0.27	0.36	0.38	0.00	100.00	0.38
Steiner's typology											
Matched ^a	12	843	0.09	0.19	40.36	-0.02	0.19	0.11	0.18	40.49	-0.18
Not matched ^a	16	948	0.03	0.17	59.49	-0.06	0.11	0.03	0.13	59.50	-0.19
Not coded ^a	6	500	0.01	0.17	42.84	-0.12	0.14	0.01	0.16	42.84	-0.25

Note. Results are corrected for predictor and criterion unreliability. *k* = number of correlations; *n* = number of teams; *SWM r* = sample-weighted mean correlation; *SWSD* = sample-weighted standard deviation of the *SWM r*; % var. due to sampling error = percentage of variance attributed to sampling error; 95% CI = 95% confidence interval; ρ = corrected population correlation; SD_{ρ} = standard deviation of the corrected population correlation; % var. acc. for = percentage of variance attributed to sampling error and artifact corrections; 95% CV_L = lower 95% credibility value.

^a To facilitate aggregation, homogeneity was used in the overall effect because the relationship between heterogeneity and performance was negative.

suggestion for future research is to use both the mean and a specified aspect of the distribution to see if the combination of the two more adequately captures team composition variables. Despite the use of simple statistical operationalizations to capture the specified composition variables, several of the team composition variable and team performance relationships were quite notable. The prospect of obtaining even stronger relationships if team composition variables are more adequately captured by a more complex operationalization is encouraging for the use of deep-level composition variables as predictors of team performance. Further, it should be noted that some of the configural operationalizations were based on only a few correlations, and thus results should be considered tentative (Hunter & Schmidt, 2004). However, the strong influence of operationalization of the team composition variable on the team composition variable and team performance relationships should serve as the impetus for future research in this area.

Some researchers have already begun to develop alternative operationalizations for team composition variables (e.g., Barry & Stewart, 1997; Kristof-Brown, Barrick, & Stevens, 2005; Lim, 2004). Although results for alternative operationalizations were included in the meta-analysis, there were relatively few studies available that operationalized team composition variables using these operationalizations. Despite this limitation, in several cases

the alternative operationalizations yielded strong relationships between the team composition variable and team performance (e.g., proportion high conscientiousness $\rho = .35$ for field teams). Given the promise of these operationalizations, researchers should include them in future studies to more accurately represent the underlying distributions of the team composition variables that are most important in predicting team performance.

One of the goals of the present study was to make theoretically and empirically based recommendations for the composition of effective teams. It should be noted these are general guidelines that may not be appropriate if there is reason to believe that implementing a specific aspect of team composition will lead to poor team performance in the particular team situation (e.g., too many agreeable or collectivistic individuals in a team that may be at risk for groupthink). Most of the recommendations are consistent with Hackman's (1987) original suggestions and more recent recommendations by Hollenbeck et al. (2004); however, many are more specific, and they are all based on the empirical results of the meta-analysis.

First, team composition can have a strong influence on team performance in organizational settings. Many of the observed effects were larger than those observed for other variables proposed to be related to team performance. Further, a few of the variables (e.g., agreeableness, conscientiousness) had effects sim-

Table 6

Meta-Analytic Results for the Relationships Between Collectivism, Preference for Teamwork, Emotional Intelligence (EI), and Team Performance

Variable	<i>k</i>	<i>n</i>	<i>SWM r</i>	<i>SWSD</i>	% var. due to sampling error	95% CI		ρ	SD_{ρ}	% var. acc. for	95% CV_L
						Lower	Upper				
Collectivism	14	1,299	0.20	0.22	21.45	0.09	0.31	0.25	0.24	22.05	-0.14
Lab	9	467	-0.01	0.13	100.00	-0.09	0.08	0.00	0.00	100.00	0.01
Field	5	832	0.29	0.20	12.90	0.12	0.47	0.35	0.22	14.49	-0.01
Mean or sum	10	1,027	0.24	0.23	16.74	0.10	0.38	0.31	0.26	17.37	-0.13
Lab	6	283	0.06	0.19	61.95	-0.09	0.21	0.07	0.15	62.01	-0.17
Field	4	744	0.31	0.20	10.76	0.11	0.51	0.40	0.24	12.09	0.00
Heterogeneity (lab only)	3	194	0.02	0.11	100.00	-0.11	0.14	0.02	0.00	100.00	0.02
Proportion high	2	120	0.10	0.08	100.00	-0.02	0.22	0.13	0.00	100.00	0.13
Steiner's typology											
Matched	2	413	0.19	0.02	100.00	0.16	0.21	0.23	0.00	100.00	0.23
Not matched	2	64	-0.05	0.13	100.00	-0.22	0.13	-0.05	0.00	100.00	-0.05
Not coded	10	822	0.22	0.26	16.52	0.06	0.39	0.27	0.29	17.06	-0.19
Preference for teamwork	10	490	0.16	0.22	41.98	0.02	0.29	0.18	0.19	42.35	-0.13
Lab	5	249	0.01	0.30	22.87	-0.26	0.27	0.01	0.32	22.87	-0.52
Field	5	241	0.19	0.14	100.00	0.07	0.30	0.22	0.00	100.00	0.22
Mean	8	380	0.20	0.17	69.42	0.08	0.32	0.23	0.11	69.95	0.06
Lab	4	166	0.18	0.23	45.04	-0.05	0.40	0.20	0.20	45.26	-0.11
Field	4	214	0.22	0.10	100.00	0.12	0.32	0.26	0.00	100.00	0.26
Heterogeneity (lab only)	2	135	0.01	0.21	32.79	-0.29	0.31	0.01	0.20	32.79	-0.32
Steiner's typology											
Matched	2	130	0.01	0.11	100.00	-0.14	0.16	0.01	0.00	100.00	0.01
Not matched	2	135	-0.03	0.18	46.29	-0.28	0.22	-0.04	0.16	46.30	-0.30
Not coded	8	360	0.24	0.13	100.00	0.15	0.33	0.29	0.00	100.00	0.29
EI (mean only)	6	304	0.16	0.12	100.00	0.06	0.26	0.18	0.00	100.00	0.18
Lab	4	258	0.17	0.09	100.00	0.09	0.26	0.20	0.00	100.00	0.20
Field	2	46	0.08	0.22	90.05	-0.22	0.39	0.10	0.08	90.12	-0.04
Steiner's typology											
Matched	2	197	0.22	0.02	100.00	0.18	0.25	0.26	0.00	100.00	0.26
Not coded	4	107	0.05	0.15	100.00	-0.10	0.20	0.06	0.00	100.00	0.06

Note. Results are corrected for predictor and criterion unreliability. *k* = number of correlations; *n* = number of teams; *SWM r* = sample-weighted mean correlation; *SWSD* = sample-weighted standard deviation of the *SWM r*; % var. due to sampling error = percentage of variance attributed to sampling error; 95% CI = 95% confidence interval; ρ = corrected population correlation; SD_{ρ} = standard deviation of the corrected population correlation; % var. acc. for = percentage of variance attributed to sampling error and artifact corrections; 95% CV_L = lower 95% credibility value.

ilar to that observed for team GMA. Second and not surprisingly, organizational teams should be composed of conscientious individuals. Third, the potential negative impact of low agreeable individuals should be considered before assigning these individuals to teams. Fourth, teams in organizations should be composed of team members high in openness to experience. The results from the meta-analysis also provide preliminary evidence for the importance of values that can benefit the team, such as collectivism and preference for teamwork. The limited evidence available in the literature suggests that composing teams with members who have a collectivistic orientation should be positively related to team performance. Similarly, organizational teams should be composed of team members who like teamwork (i.e., have a preference for teamwork). If possible, individuals who dislike teamwork or prefer to work autonomously should be allowed to self-select out of the team.

Limitations

A limitation of the present study was the unavailability of data for testing some of the relationships across different levels of the

moderators. As is to be expected with teams, the relationships between team composition variables and team performance were often complex; several moderators appear to be simultaneously affecting those relationships. Although the influence of several moderators was tested, and in most cases a large amount of variance could be attributed to artifacts, some estimates should not be interpreted as population estimates because of a limited number of correlations on which they are based. However, by including these estimates, this meta-analysis clearly indicated areas in need of future research. Specifically, more research investigating intellectual teams in field settings and physical teams in lab settings is needed. Investigations of some of the operationalizations other than the mean in lab settings are warranted. In addition, there were few correlations for the relationship between collectivism, preference for teamwork, EI, and team performance. Given that these variables show potential for at least small to medium relationships with team performance, future research is warranted. For example, although the relationship between team mean collectivism and team performance surpassed a medium effect ($\rho = .40$), there was variability in the effect ($SD_{\rho} = .24$), indicating the presence of

Table 7

Meta-Analytic Results for the Relationship Between General Mental Ability and Team Performance

Variable	<i>k</i>	<i>n</i>	<i>SWM r</i>	<i>SWSD</i>	% var. due to sampling error	95% CI		ρ	<i>SD_ρ</i>	% var. acc. for	95% <i>CV_L</i>
						Lower	Upper				
General mental ability	42	2,995	0.23	0.19	35.28	0.17	0.29	0.27	0.18	36.12	−0.02
Lab	34	2,257	0.26	0.19	35.60	0.19	0.32	0.31	0.18	36.63	0.01
Intellectual	25	1,776	0.24	0.19	35.27	0.16	0.31	0.28	0.18	36.17	−0.01
Physical	2	96	0.25	0.08	100.00	0.14	0.36	0.30	0.00	100.00	0.30
Other/mixed	7	385	0.36	0.20	35.91	0.21	0.51	0.44	0.19	38.28	0.13
Field	8	738	0.14	0.15	46.56	0.04	0.25	0.18	0.13	47.23	−0.04
Intellectual	1	79	0.35								
Physical	7	659	0.12	0.14	53.24	0.02	0.22	0.15	0.12	53.77	−0.04
Mean and sum	38	2,621	0.26	0.21	29.54	0.20	0.32	0.31	0.21	30.52	−0.02
Lab	32	2,170	0.27	0.21	29.35	0.20	0.35	0.33	0.21	30.41	−0.01
Field	6	451	0.21	0.19	32.81	0.06	0.37	0.26	0.19	33.71	−0.06
Maximum	8	622	0.22	0.22	24.40	0.07	0.37	0.27	0.23	25.17	−0.11
Lab	4	404	0.35	0.16	30.14	0.19	0.51	0.42	0.16	33.80	0.17
Field	4	218	−0.02	0.04	100.00	−0.06	0.02	−0.02	0.00	100.00	−0.02
Minimum	9	716	0.28	0.22	21.99	0.13	0.42	0.34	0.24	23.17	−0.06
Lab	5	469	0.39	0.11	66.22	0.30	0.49	0.48	0.06	76.21	0.37
Field	4	247	0.06	0.22	33.92	−0.16	0.27	0.07	0.22	33.98	−0.29
Heterogeneity	9	726	0.01	0.12	89.57	−0.07	0.09	0.01	0.05	89.58	−0.06
Lab	4	390	−0.03	0.09	100.00	−0.12	0.06	−0.04	0.00	100.00	−0.04
Field	5	336	0.06	0.13	93.81	−0.05	0.17	0.07	0.04	93.98	0.01
Other (field only)	2	100	−0.04	0.01	100.00	−0.05	−0.03	−0.05	0.00	100.00	−0.05
Steiner's typology											
Matched	15	1,087	0.23	0.20	34.82	0.14	0.33	0.28	0.19	35.68	−0.02
Not matched	20	1,543	0.24	0.18	35.39	0.16	0.32	0.29	0.18	36.39	−0.06
Not coded	13	1,044	0.22	0.19	30.21	0.11	0.33	0.27	0.20	31.13	−0.05

Note. Results are corrected for predictor and criterion unreliability. *k* = number of correlations; *n* = number of teams; *SWM r* = sample-weighted mean correlation; *SWSD* = sample-weighted standard deviation of the *SWM r*; % var. due to sampling error = percentage of variance attributed to sampling error; 95% CI = 95% confidence interval; ρ = corrected population correlation; *SD_ρ* = standard deviation of the corrected population correlation; % var. acc. for = percentage of variance attributed to sampling error and artifact corrections; 95% *CV_L* = lower 95% credibility value.

additional moderators. This is consistent with a recent review of the influence of collective orientation in teams (Alavi & McCormick, 2004), which suggested that the relationship between collective orientation and team outcomes may be fairly complex.

A second limitation was that the majority of the primary studies used correlational designs (77.27%). Even in those studies that used experimental or quasi-experimental designs, the team composition variable of interest was rarely the manipulated or grouped variable. This limits the casual conclusions that can be drawn. However, all studies collected team performance data at the same time (concurrent design) or after (predictive design) the team composition variable data, offering some temporal support for causality. On average, team performance information was collected 23.44 days (*SD* = 36.05 days) after the team composition variable.

Conclusion

In summary, the present study sought to investigate the relationship between specified deep-level team composition variables and team performance. Many of the team composition variables showed promise as a means of increasing team performance. Study setting was a notable moderator in that the strength of the team composition variable and team performance relationships was dependent on the study setting (lab or field). In field studies, several

variables (e.g., agreeableness, conscientiousness, openness to experience, preference for teamwork) had moderate relationships with team performance. In contrast, the relationships between personality factors and team performance as well as values and team performance in lab settings were negligible. For the GMA and team performance and EI and team performance relationships, stronger effects were observed in lab compared with field settings. However, even when operationalization and study setting were considered as moderators, only a small amount of variance was attributed to artifacts in most of the GMA and team performance relationships, suggesting the presence of additional moderators beyond those studied here. Researchers can use the results of the meta-analysis to guide future research, and practitioners can use the results to help effectively compose teams in organizations.

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